

2.1/2.2 Adding and Subtracting Rational Expressions – Worksheet

MCR3U

1) Simplify and state any restrictions

a) $\frac{x+1}{18} + \frac{x-1}{45}$

b) $\frac{2}{3x} - \frac{1}{4x}$

c) $\frac{3}{ab} + \frac{5}{4b}$

d) $\frac{2+a}{a^2b} + \frac{4-a}{3ab^2}$

2) Simplify and state the restrictions.

a) $\frac{1}{x-6} - \frac{1}{x+6}$

b) $\frac{12}{x+8} + \frac{3}{x-9}$

$$\text{c)} \frac{x+10}{x-6} - \frac{x-3}{x+4}$$

$$\text{d)} \frac{x+5}{x+1} + \frac{x+2}{x-2}$$

3) Simplify and state the restrictions.

$$\text{a)} \frac{x}{x^2-9x+8} + \frac{2}{x-8}$$

$$\text{b)} \frac{x+3}{x+5} + \frac{x+2}{x^2+3x-10}$$

$$\text{c)} \frac{x}{x^2+3x+2} - \frac{3x-2}{x^2+8x+7}$$

$$\text{d)} \frac{x+4}{x^2-121} - \frac{2x-1}{x^2+8x-33}$$

4) Binomial expressions can differ by a factor of -1. Factor -1 from one of the denominators to identify the common denominator. Then, simplify each expression and state the restrictions.

a) $\frac{1}{x-2} - \frac{1}{2-x}$

b) $\frac{2b+3}{4b-1} + \frac{b+6}{1-4b}$